Report labels as supervision: our solution to the RSNA 2026 Knee Abnormality Detection challenge

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Kaggle, 30 July – 22 October 2026 · private LB 0.942, rank 771 of 6,425

Abstract

The challenge asks for twelve independent findings per knee MRI study, scored by macro ROC-AUC, but only 58 of the 4,407 training studies carry labels. Every training study has a free-text radiology report, and the test studies have none, so we treated the task as weak supervision. Training targets mix three public LLM readings of the reports, half and half, with a quantile-matched image teacher. Our models read each study as windows of three adjacent slices drawn from six standardised series, encode every window with a 2D CNN, and pool the windows separately for each finding with an attention head. Our best single model, a ConvNeXt-T, scored 0.942 on the public leaderboard and 0.928 on the private one; our best ensemble of our own models scored 0.945 and 0.932. Our two final picks recombined public checkpoints and scored 0.958 public and 0.942 private, rank 771 of 6,425. All 109 of our scored entries lost about 0.015 from public to private, our own models and the public blends alike, and the order of our entries held (Spearman 0.988).

0.942	771 / 6,425	0.932	0.928
private LB, final picks C19 and C14	private rank (top 12.0 %); public rank 570	private, best ensemble of our own models (public 0.945)	private, best single model, ConvNeXt-T (public 0.942)

1 Task and metric

The training set has 4,407 knee MRI studies with 24,371 series: 3 to 14 per study, median 5, in sagittal, coronal and axial planes. Each study must receive twelve scores: ACL, MCL, medial and lateral meniscus, medial, lateral and patellofemoral osteoarthritis, effusion, synovitis, Baker's cyst, contusion and fracture. The score is the unweighted mean of the twelve per-finding AUCs,

$$\mathrm{AUC}_{\mathrm{macro}} = \frac{1}{12}\sum_{k=1}^{12} \mathrm{AUC}(\{\hat{y}_{ik}\}_i, \{y_{ik}\}_i) \tag{1}$$

Three facts shaped the design. Only 58 studies have the twelve labels, so the labelled set is a noisy validation set and nothing more; its per-finding Hanley–McNeil standard error is about 0.09. Every training study has a report, in roughly ten languages, and the test studies have none, so text can only enter through the training targets. And AUC reads only rank order: calibration is worth nothing, ensembles should average ranks rather than probabilities, and a rare finding counts as much as a common one.

2 System overview

Figure 1 shows the two lanes. In the label lane, reports become probabilities through three public LLM label tables, which we average and mix with the predictions of a public image teacher. In the image lane, the DICOM series are converted once into a compact slice cache. Our models train on the cache against the

mixed targets; the 58 labelled studies only validate. A per-finding rank-mean combines our members. The figure also separates what we submitted: our own ensemble, forks that blended it into public notebooks, and the two final picks, which recombined public checkpoints only.

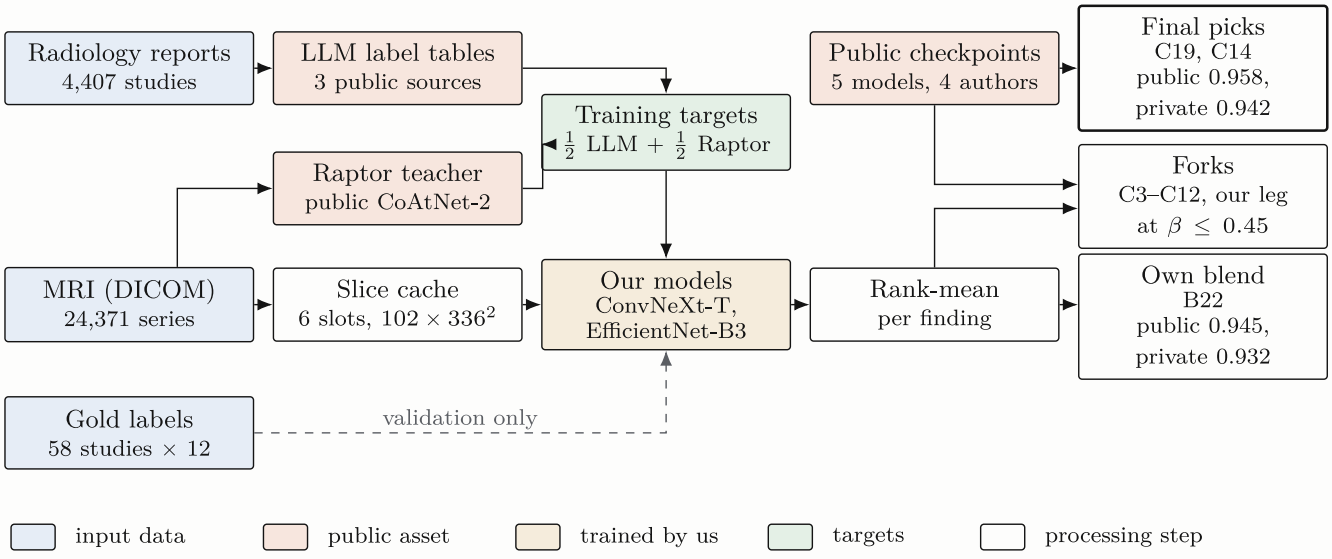


Figure 1: End-to-end view. Colours encode the role of each block (legend). Scores are public / private macro AUC.

3 Preprocessing

Series metadata as delivered is partly unusable: the two sequence flags occur only as (1, 1) or (0, 0) across all 24,371 series. We therefore re-derive every series' type from the DICOM headers: the plane from `Anatomical_Plane`, falling back to the orientation vectors; the weighting from repetition and echo time (TR $\leq$ 800 ms is T1; otherwise TE $\geq$ 60 ms is T2, else PD); and fat suppression from the series description. Each study then fills six fixed slots (Figure 2), preferring the series whose slice count is nearest 32, so that the same slot holds the same kind of image in every study.

Slices are ordered by their position along the slice normal, never by file name; on this data the file order correlates with the true order at $\rho = -0.012$. Each slot samples its slice budget equidistantly inside the 2–98 % band of the stack. A 150 mm physical crop, a per-series 1st–99th percentile intensity window and a mirror for right knees make the images comparable across scanners and sides. The result is one array of 102 slices of 336 $\times$ 336 pixels per study, stored as uint8 with a presence mask for missing slots: about 51 GB for the training set, built once on Kaggle CPU kernels.

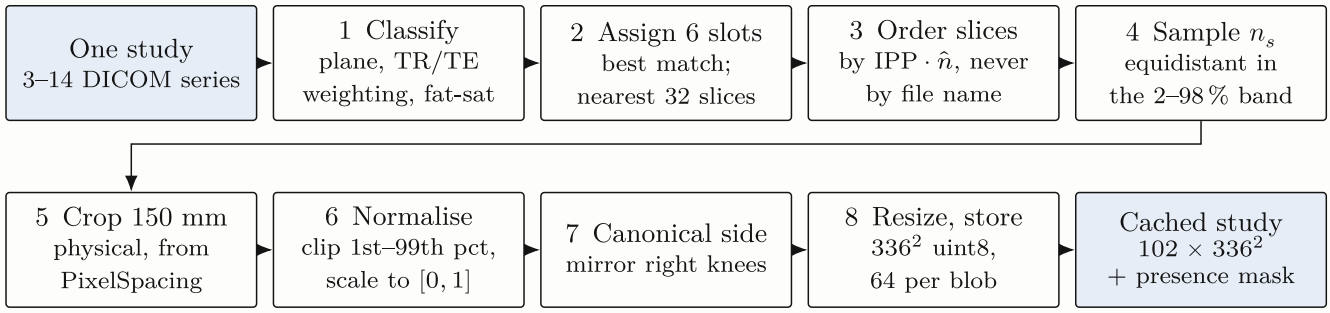


Figure 2: Preprocessing. Steps 1–4 decide which slices represent the study; steps 5–8 make them comparable. The same function runs on the hidden test studies at inference.

4 Training targets

The 4,349 unlabelled studies get soft targets from two sources. The first is the mean of three public, CC0 LLM label tables, which read each report into twelve probabilities, $\bar{p}_{ik}$. On the 58 labelled studies this blend reaches a macro AUC of 0.8948. The second is an image teacher: the Raptor branch of a public notebook, three CoAtNet-2 checkpoints over four views, which we ran over all training studies in three Kaggle shards. Its raw outputs r_{ik} have a different scale, so we map each one to the LLM blend's quantile at the same rank before mixing (Figure 3):

$$q_{ik} = \hat{F}^{-1}_{\bar{p},k}\left(\frac{\mathrm{rank}_k(r_{ik}) - \frac{1}{2}}{n}\right), \qquad y^{\mathrm{t}}_{ik} = (1-\lambda)\,\bar{p}_{ik} + \lambda\, q_{ik}, \quad \lambda = 0.5 \tag{2}$$

Each cell also carries a confidence weight that is high when the three LLM sources agree and are far from 0.5, with σ_{ik} their standard deviation:

$$c_{ik} = \mathrm{clip}\Big(\tfrac{1}{2}(1 - 2\,\sigma_{ik}) + \tfrac{1}{2}\big|2\bar{p}_{ik} - 1\big|,\ 0.15,\ 1\Big) \tag{3}$$

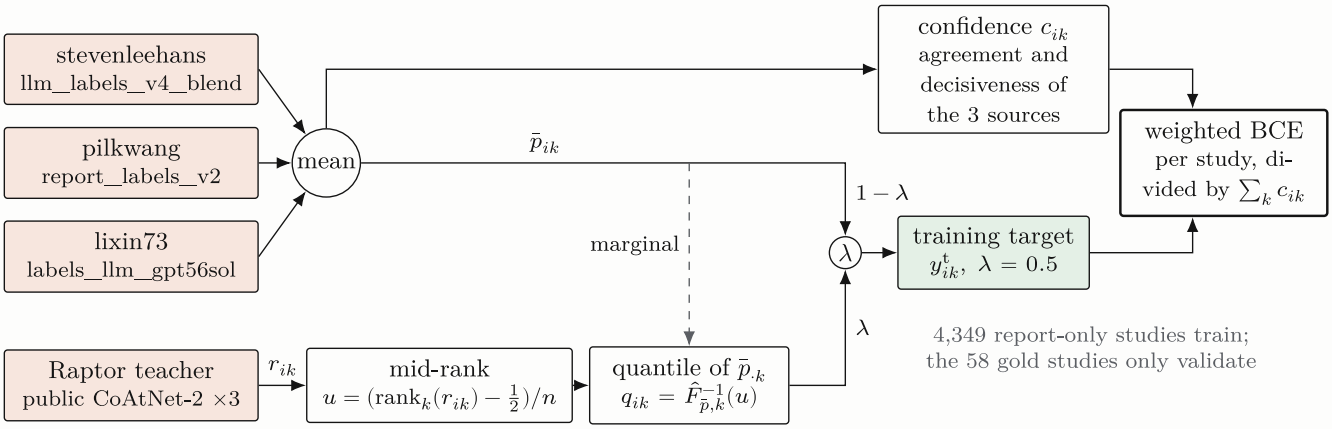


Figure 3: Training targets for study i and finding k. Quantile matching keeps the LLM blend's marginal distribution and takes only the teacher's ordering.

This mix was the only target change that moved the leaderboard. The same CoAtNet-1 recipe scored 0.918 on LLM targets and 0.927 with the Raptor half (submissions 18 and 20; private 0.902 and 0.914). Self-distillation, a second teacher, a cross-fitted out-of-fold student, a fresh relabelling of all reports with a hosted LLM, report-only targets and a share of a fourth public label table all read flat or lower (Table 3).

5 Model

A study is a variable set of 2.5D windows. A window $w = (s,\ c)$ takes slices $c - 1$, c and $c + 1$ of slot s as the three colour channels of one 288×288 image, for every interior slice of every present slot: at most 90 windows per study. One ImageNet-pretrained 2D encoder maps each window to a feature vector h_w; all windows of a batch go through it in a single pass (Figure 4a).

The head (Figure 4b) adds a learned slot embedding and layer normalisation, then scores every window for every finding with a small gate MLP g. A softmax over windows, with absent windows masked ($m_w = 0$), gives each finding its own attention distribution, and a per-finding linear readout gives the logit:

$$a_{\ell w} = \frac{m_w \exp g_\ell(\tilde{h}_w)}{\sum_{w'} m_{w'} \exp g_\ell(\tilde{h}_{w'})}, \qquad z_\ell = \Big\langle \sum_w a_{\ell w}\, \tilde{h}_w,\ v_\ell \Big\rangle + b_\ell \tag{4}$$

The per-finding attention matters here because the twelve findings live in different places: a meniscal tear is visible on a few sagittal and coronal slices, an effusion on many axial ones. Each finding can choose its own windows instead of sharing one pooled study vector.

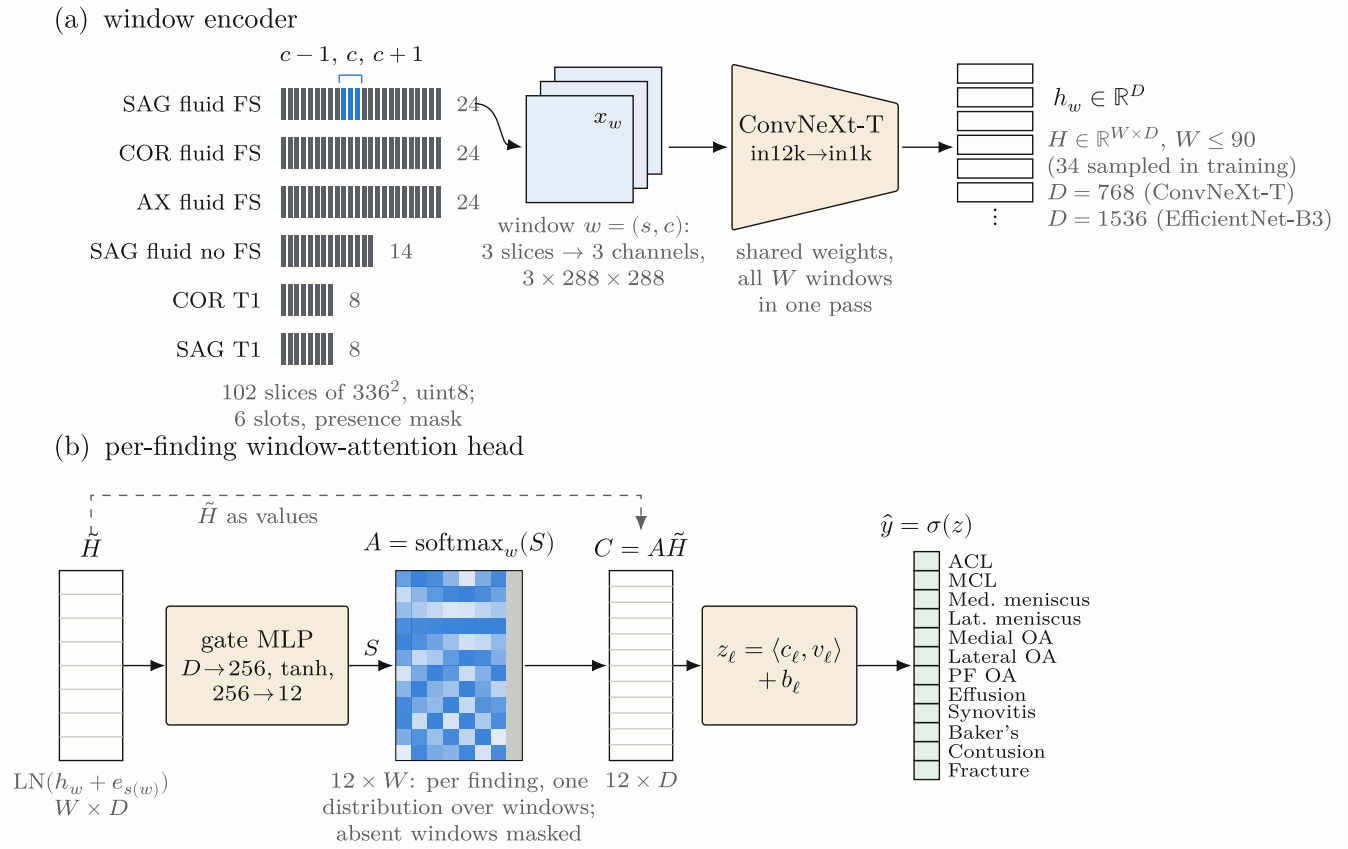


Figure 4: The production architecture. (a) Window encoder: every window of every slot is one image for a shared 2D CNN. (b) Per-finding window-attention head: each finding attends over all windows of the study (the attention values shown are illustrative). Shapes are for ConvNeXt-T.

Table 1: The two production members. Both use the architecture of Figure 4, the same data and targets, and differ in the encoder and its optimiser.

	`v15c`	`v13b3`
encoder	ConvNeXt-T, in12k → in1k	EfficientNet-B3, ra2
feature size D	768	1,536
encoder learning rate	1e-4, × 0.9 per stage	3e-4, uniform
normalisation	LayerNorm (no BN)	BatchNorm, statistics frozen
epochs; checkpoint	20; mean of EMA, epochs 17–19	30; mean of EMA, epochs 27–29
training time	1.5 h, one RTX 4090	2.2 h, one RTX 4090
public / private, alone	0.942 / 0.928	0.940 / 0.926

6 Training

Every model trains on all 4,349 report-only studies and validates on the 58 labelled ones; there is no fold split and no epoch selection. The loss is binary cross-entropy against the soft targets of equation (2), weighted by the confidences of equation (3) and normalised per study:

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^{B} \frac{\sum_k c_{ik}\, \mathrm{BCE}(z_{ik},\, y^{\mathrm{t}}_{ik})}{\sum_k c_{ik}} \tag{5}$$

Shared settings: AdamW with weight decay 0.02 (none on biases and 1-D parameters), head learning rate 1e-3, 10 % linear warm-up then cosine decay, gradient clipping at 1.0, fp16 mixed precision, an exponential moving average of the weights (decay 0.998), two studies per step with two-step accumulation, 34 windows sampled per study, drop-path 0.1. Augmentation is applied per window with probability 0.9: rotation ±15°, zoom 0.90–1.15, shift ±8 %, gamma 0.7–1.4, contrast 0.8–1.25, gain 0.85–1.15, cut-out with probability 0.3, and Gaussian noise; never flips, because the side is already canonical. The final checkpoint is the mean of the EMA snapshots of the last three epochs.

The one lesson that cost us most to learn: an encoder family needs its own optimiser. CNNs that shared the transformer recipe (small learning rate, layer-wise decay, trainable BatchNorm) under-performed until they got a uniform 3e-4 with frozen BatchNorm statistics, which lifted a ResNet-34 from 0.831 to 0.899 on the labelled studies; ConvNeXt-T, in turn, only reached 0.942 with its own stage-wise decay.

7 Ensembling and the final submission

7.1 Rank-mean of our members

Because the metric reads only order, members are combined per finding by their percentile ranks π:

$$s_{ik} = \sum_m \omega_{mk}\, \pi_{ik}^{(m)}, \qquad \pi_{ik}^{(m)} = \frac{\mathrm{rank}_k\big(\hat{y}_{ik}^{(m)}\big)}{N}, \qquad \sum_m \omega_{mk} = 1 \tag{6}$$

Our own blends used flat weights. The gain over the members' mean grew with the number of different families: on the labelled studies, pairs from different families gained +0.0053 on average and pairs of the same recipe +0.0025. ConvNeXt-T plus EfficientNet-B3 (B18) scored 0.944 public / 0.930 private; adding a ConvNeXt-T trained on report-only targets as a third vote (B22) gave 0.945 / 0.932, our best own result.

7.2 Forks of public notebooks

Public notebooks moved the leaderboard plateau from 0.942 to 0.958 during the competition, mostly through newly published trained models. We blended our ensemble into these notebooks at a weight β and found a stable rule:

$$\mathrm{LB}(\beta) \approx (1-\beta)\,\mathrm{LB}_{\mathrm{anchor}} + \beta\,\mathrm{LB}_{\mathrm{leg}} + \kappa\,\beta(1-\beta) \tag{7}$$

where the diversity term κβ(1 − β) was worth +0.002 to +0.004 on public. It held on private: heliosli's ConvNeXt-S (0.930 private alone) plus our B18 (0.930) gave 0.937, above both. On the strongest anchors, however, our leg was about 0.012 weaker than the anchor and cost a tick on public even at $\beta = 0.05$.

7.3 The final picks

Our final picks therefore contain no weights we trained. Both start from heliosli's public “0.9586” notebook (V2), which blends a Raptor half of two CoAtNets with a compact half built on a ConvNeXt-S, per finding and by rank. C14 adds nartaa's s6fall ConvNeXt-S to the compact half at 0.4; C19 is C14 with the four per-finding Raptor-half weights of heliosli's earlier 0.958 notebook in place of V2's (Figure 5). Both read 0.958 on public; we ordered them by our place inside the tie of about 1,500 teams at 0.958, which only shows the sign

of a sub-tick difference. The two CoAtNets were trained with the OAI knee dataset as extra labels. The host never ruled on OAI by name, and we accepted that rule risk.

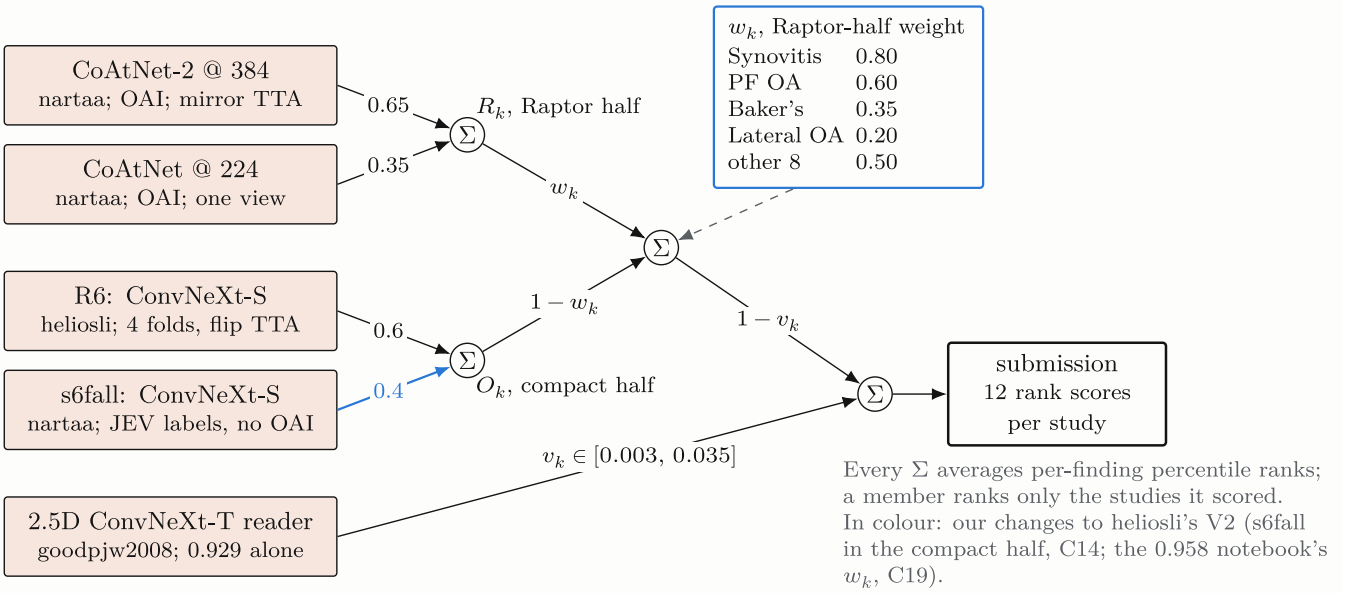


Figure 5: Structure of the final pick C19. Each Σ is a weighted average of per-finding percentile ranks, equation (6); edge labels are the weights.

8 Results

We made 109 scored submissions between 28 August and 21 October (Figure 6). Private scores were revealed after the deadline, so every entry below now has both. The early climb came from the input (the slice cache, +0.030), from blending two heads on one backbone (+0.019) and from the Raptor half of the targets (+0.009). The last own-model gains came from the CNN recipe and from ConvNeXt-T on its own optimiser.

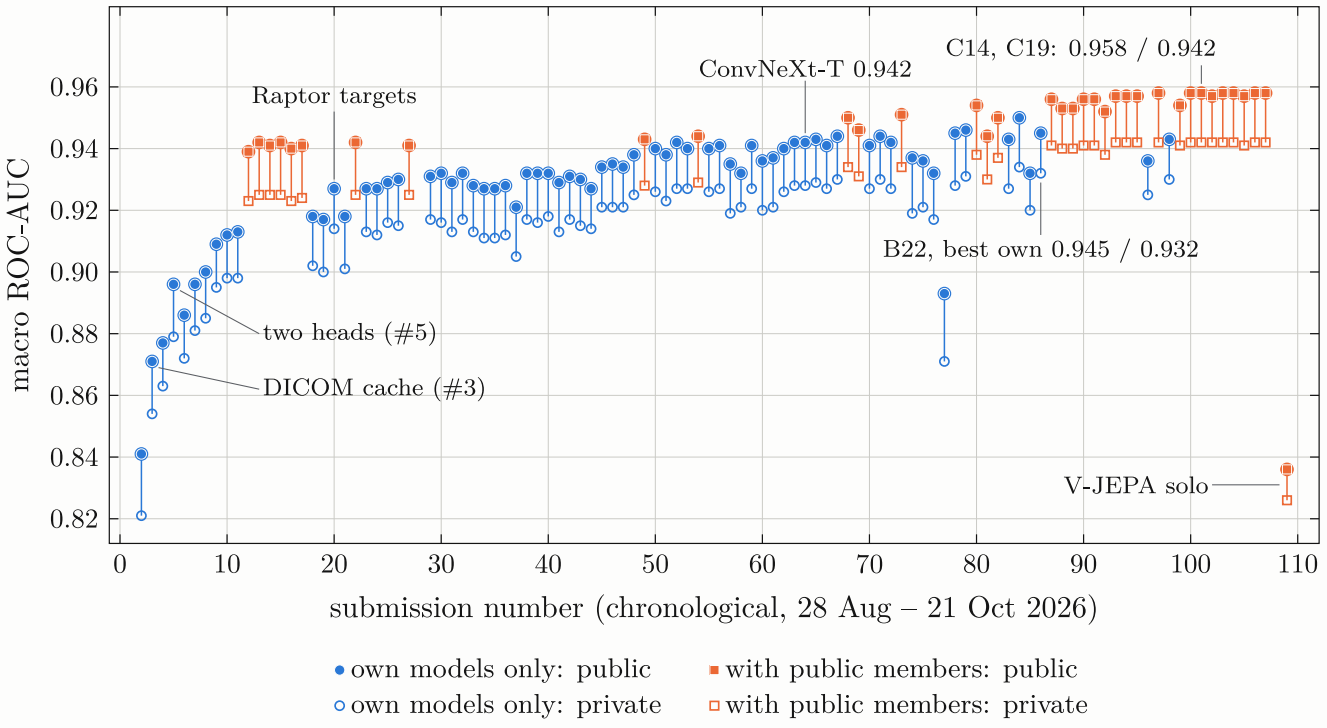


Figure 6: All scored submissions in order. Each vertical segment runs from the public score down to the private score of the same entry. Not shown: a smoke test (#1, 0.500) and a deliberate per-finding probe (#28, 0.785). Rounded to three decimals by Kaggle.

Table 2: Selected entries. "Own" entries contain only models we trained; "fork" entries blend our models into a public notebook; "public" entries recombine public checkpoints.

#	entry	kind	public	private
2	DINOv2-S, first real model	own	0.841	0.821
3	+ slice cache (crop, laterality, normalisation)	own	0.871	0.854
5	two heads on one backbone, rank-mean	own	0.896	0.879
11	seven-model blend	own	0.913	0.898
18	CoAtNet-1, LLM targets	own	0.918	0.902
20	same, ½ LLM + ½ Raptor targets	own	0.927	0.914
50	EfficientNet-B3 (`v13b3`)	own	0.940	0.926
64	ConvNeXt-T (`v15c`), best single	own	0.942	0.928
67	B18 = `v15c` + `v13b3`	own	0.944	0.930
86	**B22 = B18 + report-only ConvNeXt-T**	**own**	**0.945**	**0.932**
15	public 0.942 stack, unchanged	public	0.942	0.925
69	0.942 stack + B18 at β 0.45	fork	0.946	0.931
82	heliosli S6 + B18	fork	0.950	0.937
99	heliosli R7 + s6fall + B18, no OAI	fork	0.954	0.941
93	R6 + flip notebook + our leg at β 0.20	fork	0.957	0.942
101	C14 = V2 + s6fall	public	0.958	0.942
106	**C19 = C14 + the 0.958 weights (pick 1)**	**public**	**0.958**	**0.942**

8.1 Public versus private

The private split was harder for every entry by almost the same amount (Figure 7). Over our 98 entries above 0.90 public, the drop averaged 0.0149 with a standard deviation of 0.0014: 0.0146 for our own models, 0.0154 for entries with public members, and 0.0157 for the heavily tuned 0.957–0.958 notebooks. Those notebooks therefore paid about one tick for their public tuning and still finished 0.010 above our best own ensemble. Their per-finding weights made no visible difference on private: the notebook before the weights (#93), with the 0.958 set (#97) and with V2's set (#100) all read 0.942.

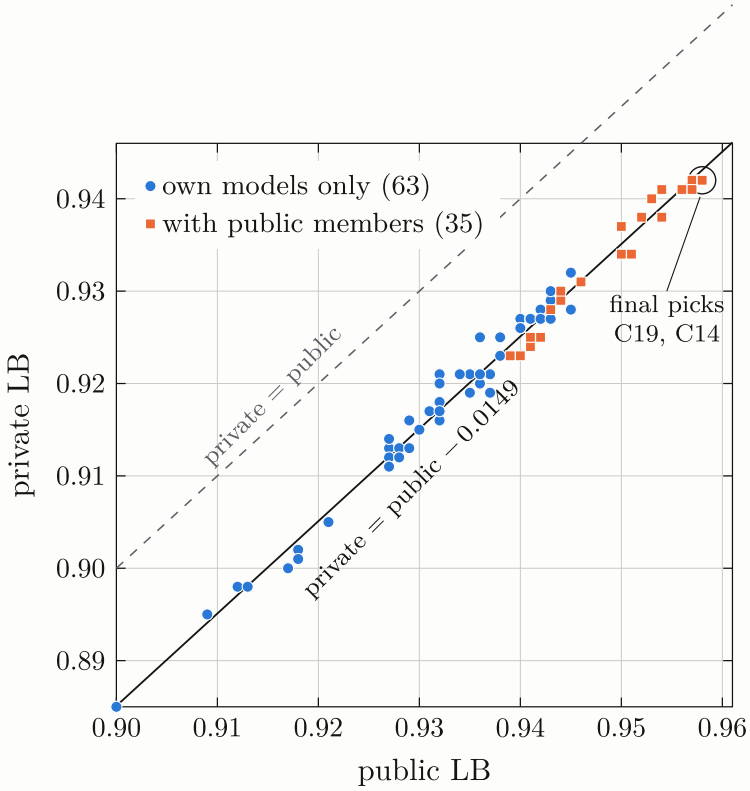


Figure 7: Private against public score for every entry above 0.90 public. The solid line is the mean drop of 0.0149; the dashed line is no drop. The public order of our entries held on private (Spearman 0.988).

8.2 Where we finished

We finished at 0.942 private, rank 771 of 6,425, after 570 on the final public board. At three decimals our eleven best entries all read 0.942, so no choice of two picks could have scored more. The 201 places were lost inside the public tie: 1,638 teams shared 0.958 on public, mostly with copies of the same notebooks, and 1,239 of them landed on 0.942. Of the 770 teams above us on private, 256 were above 0.958 on public (all of them stayed above us), 290 were above us inside the tie already, 179 passed us from below inside the tie and 45 came from 0.957 or lower; 23 teams fell from above us to below. The teams that held their places had their own models: the 256 public leaders averaged 0.9475 on private.

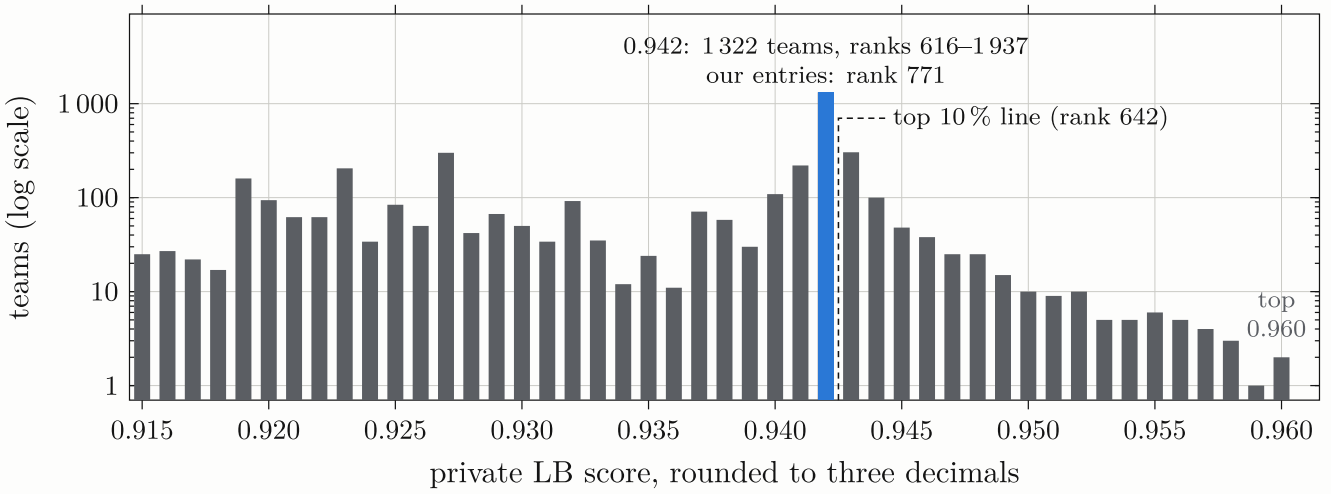


Figure 8: The private leaderboard. The spike at 0.942 is the public 0.958 tie after the shake-up; ties are ordered by unrounded score, which places us 156th of the 1,322 teams there.

9 What did not transfer

Most of our sixty-odd experiments read flat. Table 3 lists the main ones next to the control they changed, with both scores now that private is known. The private column agrees with the public verdicts; in particular, OAI labels added +0.003 on public and nothing on private.

Table 3: Ideas that did not transfer, each against its control (public / private).

change	with	control
self-distillation targets (CoAtNet-1)	0.917 / 0.900	0.918 / 0.902
second teacher, D4 CoAtNet (c03 CoAtNet-1)	0.930 / 0.915	0.932 / 0.916
five cross-fitted folds instead of one model	0.928 / 0.913	0.927 / 0.914
per-finding spatial reader head	0.929 / 0.913	0.932 / 0.916
all reports relabelled with a hosted LLM (EfficientNet-B0)	0.935 / 0.919	0.935 / 0.921
cross-family out-of-fold student (EfficientNet-B0)	0.937 / 0.919	0.935 / 0.921
DINOv2-S on the CNN data recipe	0.932 / 0.917	0.942 / 0.928
report-only targets (ConvNeXt-T)	0.932 / 0.920	0.942 / 0.928
OpenJev labels inside the teacher half (ConvNeXt-T)	0.936 / 0.925	0.942 / 0.928
OAI knees as external labels (ConvNeXt-T)	0.945 / 0.928	0.942 / 0.928
scoring at 320 px instead of 288 px	0.942 / 0.927	0.942 / 0.928

Two more directions closed late: DICOM site and protocol metadata predict the reports' labels but not the image-read ones, and a site prior or per-site centring lowered the labelled-set AUC; and the only new public model family late in the competition, a V-JEPA 2.1 ViT-B, scored 0.836 public and 0.826 private on its own after we fixed a loader bug that had blanked every series longer than 24 slices.

10 Lessons

- **Input and targets before the encoder.** The largest single gains came from the slice cache (+0.030) and the Raptor half of the targets (+0.009); after that, every target change we tried was flat.
- **A family needs its own optimiser.** Moving a CNN onto its own recipe was worth more than any architecture change.
- **Diversity pays, in proportion to how different the members are.** Different families gained about twice what seeds of one recipe gained, and the diversity gain survived the private split.
- **58 labels cannot decide between close variants.** The labelled set reversed the leaderboard direction several times; we judged members by their solo leaderboard score instead.
- **A public plateau is a lottery inside, not a cliff.** The tuned public blends lost one tick more than our own models, yet the order inside a 1,638-team tie decided 201 places. Only an own model clearly above the plateau would have kept them.

All scores are Kaggle's, rounded to three decimals; the private board was read on 23 October 2026 before Kaggle verified it. Code: `src/kaggle_pipeline.py` (training and inference), `src/cache_pipeline.py` (slice cache), `src/build_targets.py` (targets), `src/build_anchor_copy.py` (final picks); every measurement with its verdict is in `docs/experiments.md`. Figures and equations were typeset with TikZ, pgfplots and XeLaTeX in Latin Modern.